

Design, fabrication, testing, and packaging of the 1 × 8 power splitter developed under the SOI platform

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Received 11 September 2024; revised 11 January 2025; accepted 14 January 2025; posted 14 January 2025; published 12 February 2025

This paper presents a brief roadmap to design and fabricate a final device corresponding to a 1 × 8 power splitter developed under the silicon-on-insulator (SOI) platform. To describe the entire technological process for the power splitter, first, we begin by designing and validating our proposed photonic circuit through calculations using several computational routines supported by popular numerical methods, such as the finite-difference time-domain (FDTD) and beam propagation methods (BPM). In the second step, the technological platform employed for fabricating the designed optical circuit is briefly described. Subsequently, the primary objective of this paper is presented, which is to provide a detailed introduction and explanation of the assembly process, including the positioning and coupling of the optical signal to the chip and the integration of all components. Additionally, we developed the packaging process, designing the case to include internal pieces and the external housing suitable for the final system, including fiber optics input/output interfaces. In this final step, all internal and cover components for the packaging were fabricated using 3D printing. © 2025 Optica Publishing Group. All rights, including for text and data mining (TDM), Artificial Intelligence (AI) training, and similar technologies, are reserved.

<https://doi.org/10.1364/AO.541307>

1. INTRODUCTION

In recent years, the silicon photonic platform has become the most popular scheme for designing and fabricating various integrated optical circuits. A major advantage of this technology is its high compatibility with the CMOS process, which enables the production of photonic components and devices with high integration levels [1,2]. Currently, this platform is central to the production of most devices used in optical communications. Moreover, this photonic topology has recently emerged as an alternative method for mass-producing a wide variety of photonic sensors and different optical micro-systems applicable in various technological fields, including biology, material science, medicine, environmental issues, and robotic control [3–6]. This scheme allows for the creation of numerous photonic architectures suitable for complex technological applications.

Key circuits associated with this technology include ring resonators, Mach-Zehnder interferometers, directional couplers, and power splitters, which are integral to many photonic designs. Furthermore, new and innovative applications are being developed in crucial technological fields, such as quantum information and computation. Highly specialized sensors are being designed for implementation in health and metrology,

alongside optical combs and communications, micro-machines, and more recently, all-optical processors that inspire new paradigms for computation [7–12].

On the other hand, a significant drawback of optical assembly in silicon photonics is the manipulation of components that are smaller than 1 μm. Addressing this challenge requires the development of a robust and automated method for assembling the input and output components of our photonic systems on the silicon platform. To overcome these fabrication challenges and promote innovative designs, it is essential to develop an effective strategy for assembling and packaging the new device.

Currently, there are no established standards for this process, making it an important area to address promptly. Achieving a suitable optical connection typically requires maintaining tight tolerances for alignment and ensuring compatibility with standard electrical system assemblies. Additionally, optical interfaces are highly sensitive to losses and propagated wavelengths [13,14].

Similarly, optical assembly requires high precision to connect the chip with other optical components like fiber arrays. New UV epoxy adhesives are frequently employed, offering performance levels suitable for aerospace compatibility. Additionally,

creating proper interfaces is crucial, making the precise fusion of fibers to the input and output (I/O) plugs a meticulous task. Fast and straightforward fiber connectors, such as SC-APC or PUC, are required for public FTTH networks to comply with optical communications standards.

Currently, integrated photonics packaging lacks a standardized approach. Laboratories, research groups, and companies develop their proprietary strategies and methods tailored to each photonic device. Each organization devises its procedures based on specific challenges they encounter [15,16].

Furthermore, when establishing passive optical interconnections, integrating laser sources and optical components (such as micro-lenses or mirrors) is essential. Engineers must consider the optical interface, thermal issues, and compatibility with the routine packaging process.

In recent years, new tools have emerged as powerful means to incorporate straightforward and cost-effective procedures for integrated photonics prototyping. This trend is gaining popularity, making these tools accessible to small laboratories and useful for teaching purposes at undergraduate and higher education levels. In this context, 3D printing is an especially useful resource, facilitating rapid prototyping for laboratory testing and impacting several development areas [12–16].

Moreover, adopting careful procedures for integrated photonics is invaluable to the optical and technological communities. The silicon photonic community strongly desires thorough descriptions of each step in these processes.

Given these considerations, this paper aims to provide readers with a detailed fabrication process for a power splitter designed to operate in the communication band. This includes design

and simulation to support optical performance on the SOI platform. Additionally, the paper presents assembly and packaging stages, highlighting the main contributions of this study, including the use of 3D printing for designing and fabricating the optical splitter's case including the internal pieces for a suitable packaging.

2. MATERIALS AND METHODS

A. Materials

The power splitters were implemented on a silicon-on-insulator (SOI) wafer, which features a silicon core with a thickness of 340 nm, overlaid by a 2- μm layer of SiO_2 . This structure is supported on a silicon substrate.

The chip fabrication processes have followed the standard UV-lithography and associated methods as described in [17]. The designed optical circuit was inspired by an innovative design based on simplified coherent coupling (SCC). In addition, a small MMI was located at the entrance to split the input propagated mode into two modes to be launched for each branch of the splitter [18]. Every design has an attached grating coupler to perform I/O light coupling. In Fig. 1(a), the proposed 1×8 power splitter is displayed. In the middle, a detailed design of the 1×2 splitter is shown. This key optical circuit serves as the foundation for each branch of the 1×8 device and is highlighted in green in Fig. 1(a) of the schematic. The development of this device followed the strategy outlined in [18]. Additionally, on the right, Fig. 1(b) presents an optical microscope image of the fabricated 1×8 splitter. Finally, Fig. 1(c) shows a schematic of the grating coupler used to couple and decouple light from the 1×8 splitter.

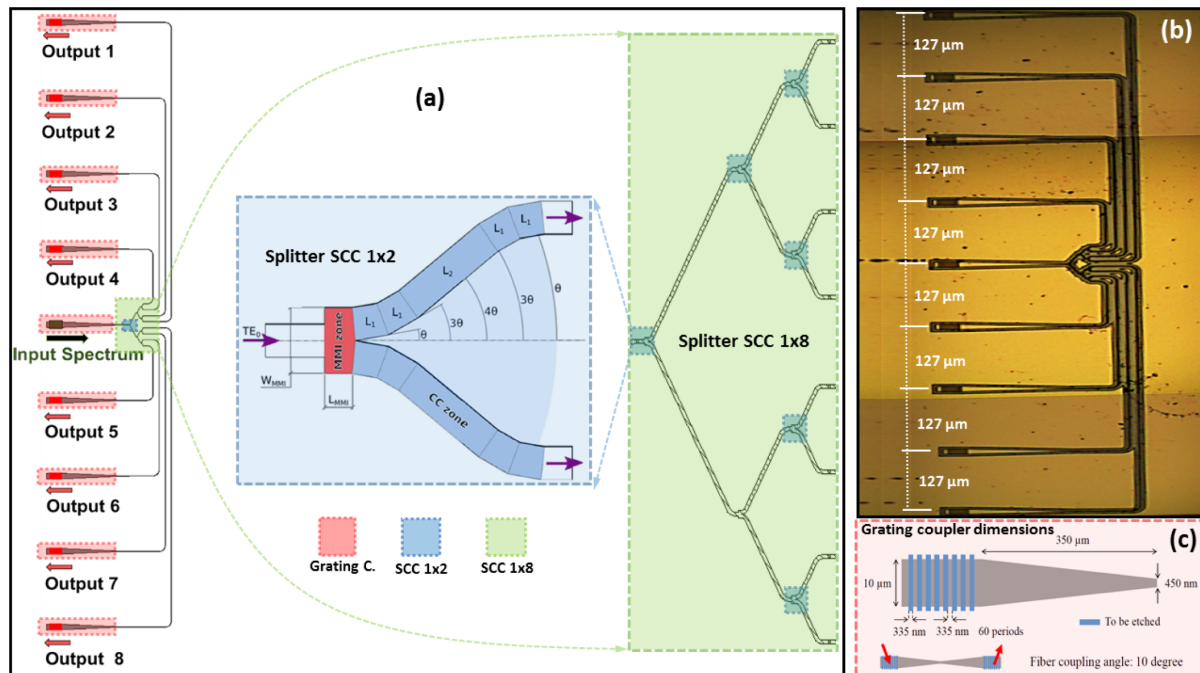


Fig. 1. (a) A schematic of the photonics circuit for the 1×8 splitter is presented. In the middle, the scheme for a 1×2 splitter, which forms the foundation of the device, is detailed [18]. As illustrated, a small MMI at the input, along with the output branches designed using the SCC strategy, underpins the design of our splitter (highlighted in green, far right). (b) An optical microscope image of the 1×8 design is shown, taken from the SOI photonics chip. (c) The layout of the grating coupler is depicted.

On the other hand, a tunable communication band laser (1530–1560 nm) from Shenzhen Box Optonics Technology (China) stepped by a 50 GHz laser was used to explore the power splitters. Initially, to ensure reproducibility to inspection by using the six-axis motorized platform, the photonic chip was fixed by adhesive to a microscope glass slide. To measure the optical power at the output we have used two systems depending on the required sensitivity; these are an InGAS head coupled to the trans-impedance amplifier and connected to a PC by a LabJack T4 card. Besides, for large sensibility, we used a Germanium power meter (PM101U - Power Meter Interface from Thorlabs Inc., USA).

B. Design and Simulation Performance

The design of the 1×2 power splitter was undertaken, and numerical simulations were subsequently performed to validate the optical performance of the proposed circuit. The splitter design is grounded in the simplified coherent coupling (SCC) theory, which enables compact optical branches and minimizes the device footprint [18]. Additionally, a small multimode interference (MMI) structure was integrated at the input to split the propagation mode from the initial straight waveguide.

Figure 2(a) presents the simulation results for the relative optical transmission as a function of the MMI length (L_{MMI}). The propagation characteristics were analyzed using the finite-difference time-domain (FDTD) method, with the key features

of the 1×2 power splitter depicted in Fig. 2(b). The optical transmission as a function of the length parameter L_{MMI} , a critical factor in the SCC design as outlined in [18], is shown in Fig. 2(c). Lastly, Fig. 2(d) provides the FDTD simulation results for the splitter across the communication band, with the inset illustrating the input and output modes of the SCC power splitter fabricated on the SOI platform. In addition, optical transmission from the 1×8 power splitter was simulated using the aforementioned software. The results from simulations are presented in Table 1 for comparison with the experimental data.

C. Fabrication of Optical Circuits of the SOI

A set of 1×2 , 1×4 , and 1×8 power splitters was fabricated using a silicon-on-insulator (SOI) wafer. The wafer consisted of a 340-nm-thick crystalline silicon core, positioned on top of a 2- μm -thick buried oxide layer and a silicon substrate. A 1- μm -thick silicon dioxide layer was deposited as the final cladding to encapsulate the entire device. The fabricated chips were configured as rectangular slabs with approximate dimensions of 15 mm $\times$ 20 mm.

It is well established that the single-mode strip waveguide width for telecommunications wavelengths (1530–1630 nm) is 450 nm. To define the optical circuits for the proposed 1×8 power splitters, standard UV lithography techniques were

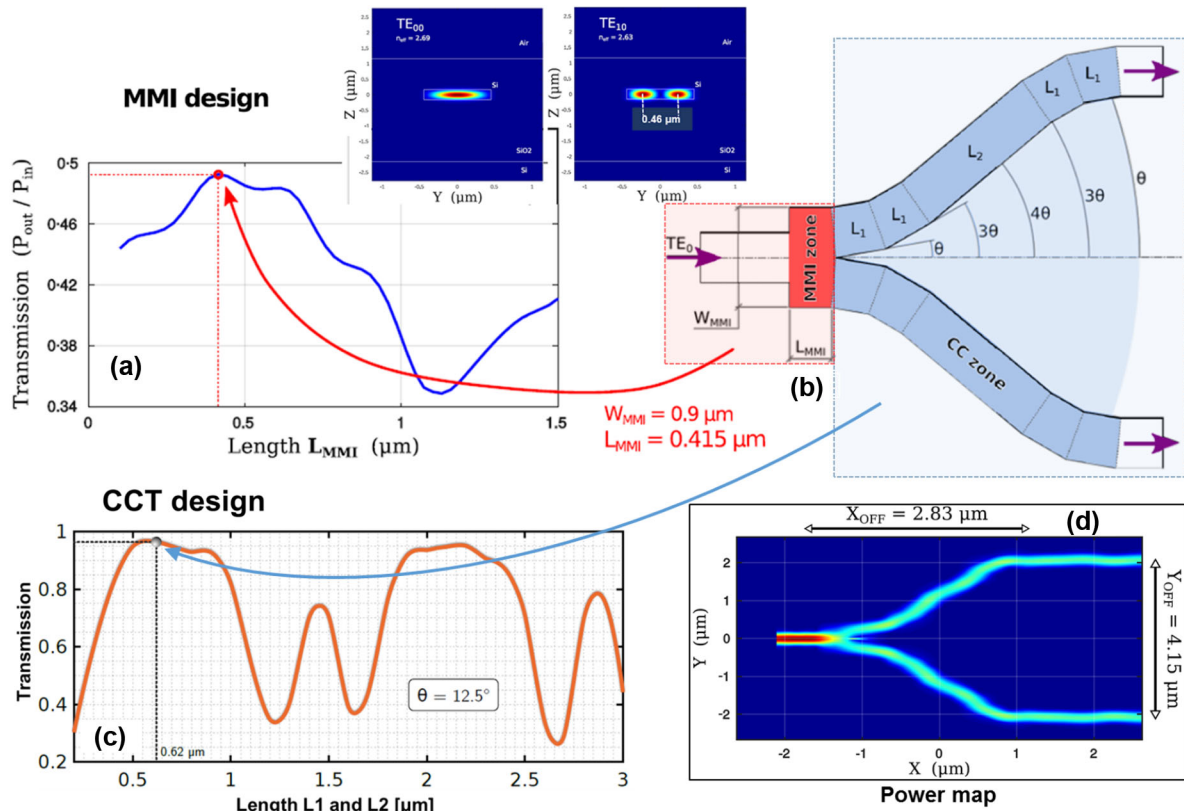


Fig. 2. Illustrates the design (b), highlighting the two main components of the 1×2 power splitter developed in this work: the MMI at the entrance and the branches designed using the SCC strategy. FDTD numerical calculations (a), (c) were conducted based on the proposed splitter design. In (a), optical transmission is plotted against the length of the MMI (L_{MMI}). (d) Presents the propagation modal distribution for the 1×2 power splitter. The insert shows the propagation modes at both the entrance and exit of the 1×2 power splitter. For further details, refer to [18].

employed, supplemented by additional fabrication methods. The entire process was conducted following stringent fabrication protocols, as described in [17,18].

D. Optical Coupling System

The coupling of light into the power splitter was achieved using a six-axis motorized station equipped with sub-micrometer resolution. The station is comprised of two main platforms: one for the X and Y stages, upon which the photonic chip is placed, and a second platform for all angular movements and the Z axis, which are meticulously aligned to position the fiber array relative to the chip. This precise positioning procedure is crucial, as it facilitates the coupling and decoupling of light for each photonic device on the chip. Figure 3 illustrates the configuration scheme of the six-axis station utilized in this study. As illustrated, the primary system is enhanced with a PC featuring dual monitors, three CCD cameras with integrated magnifying optics, and a beam profile analyzer. The optical microscopes are equipped with tailored optics to ensure unobstructed observation of the coupling region, providing both frontal and lateral views of the chip. Notably, the optical supports and magnifiers were fabricated using 3D printing, a choice that significantly reduced system setup time and overall costs.

The coupling fiber array (FA) is based on a V-groove 1×8 fiber configuration supplied by OZ-Optics (Canada).

In summary, Fig. 3 presents several perspectives and zoomed images of the described experimental setup.

A control system was developed to manage the stepper motors of the six-axis platform. This system is centered around an Arduino ESP32 microcontroller board and utilizes six DRV8825 stepper motor drivers to manage the current of each motor. The firmware was implemented using C++ within the Arduino IDE. Communication between the motor controller and a PC is established via a USB port, allowing serial ASCII commands to be sent by users for operations such as absolute, relative, and home position movements, among other functionalities. The firmware is capable of synchronizing movements across multiple motors.

Additionally, the strategy illustrated in Fig. 4 is based on the images obtained from the optical microscopes included in the experimental setup. The images present a frontal (upper) view and a lateral (lower) view of the chip array, which serves as a crucial factor in making decisions regarding the determination of the initial point for the pre-alignment process.

From Fig. 4, it can be observed that in the frontal image (upper), the array is aligned tightly perpendicular to the plane of the chip, using the border as a reference line. In contrast, the lateral view (bottom) reveals two yellow lines that guide the observation of the elevation angle, a sensitive parameter for aligning the fiber array to the chip. The chip positioning, as depicted in this image, is confirmed to be correct. Additionally, this image allows for the appreciation of the typical angle of approximately 10° , which is necessary for achieving suitable coupling between the fiber array and the silicon chip.

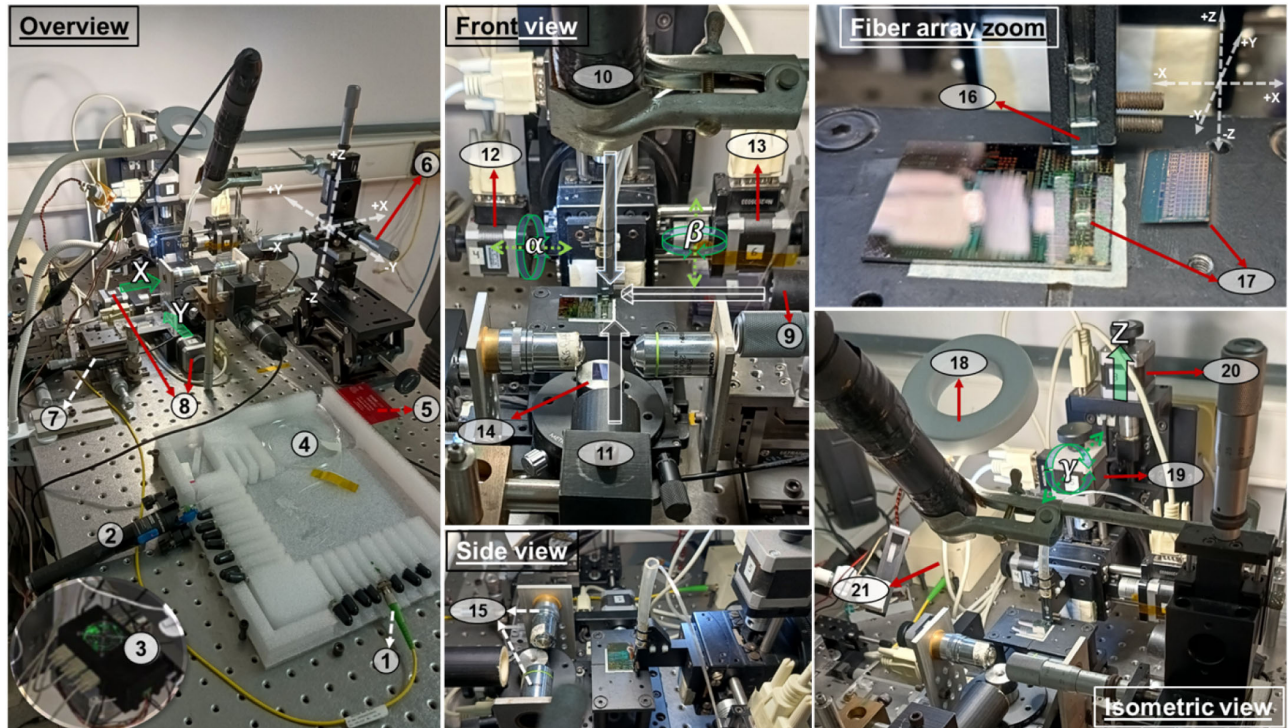


Fig. 3. Experimental setup used to couple the light into the silicon photonics structures. The six-axis motorized station (7), (8) for X, and Y planar chip movements, (12), (13), (19) for α , β , γ angular array fiber movements, and (20) for Z height displacement of the fiber array. The optical microscopes (9), (10), and (11) are the main components to support this system. In addition, by using this setup, we can manipulate the light coupling by the fiber array (16) to the chip (17). This figure also includes other components used in this work: (1) input fiber, (2) visible laser for alignment, (3) motor controller, (4) optical fibers, (5) power meter, (6) microscope positioners, (7) microscope objective positioners and (14) rotary base adapted to X, Y motors for edge coupling, (15) microscope objectives, (18) diffused light ring, (21) tunable laser in C band.

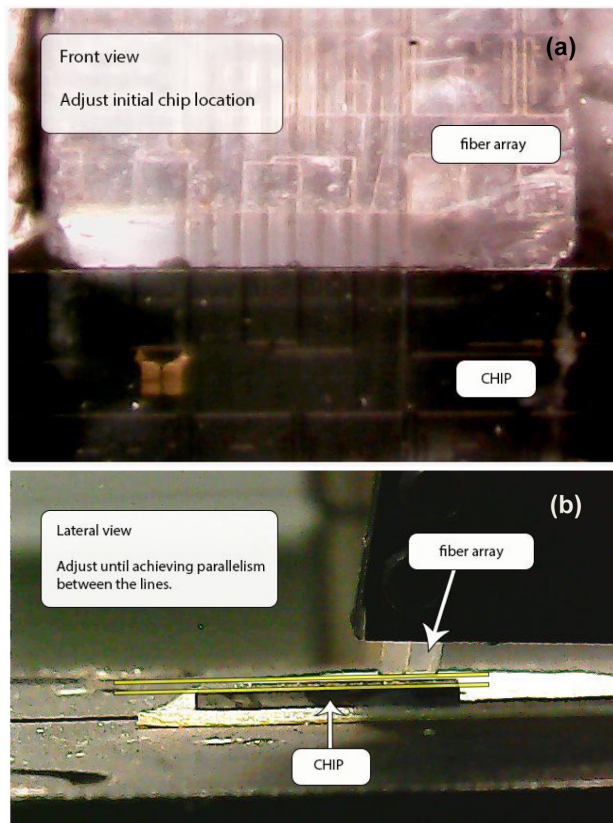


Fig. 4. Front (a) and lateral (b) views taken from two microscopes included in the experimental setup to couple light in photonics chips.

3. RESULTS AND DISCUSSION

A. Assembly and Packaging Process

1. Pre-alignment Process

Initially, the system comprising the chip and fiber array (FA) was positioned using three optically magnified CCD cameras, taking into account the border and the layout of each device in the photonic chip. These optical microscopes provided various perspectives, including frontal and lateral views. The FA was initially placed near a chip corner, which was affixed to a microscope slide and secured to the motorized station, marking the starting point for the pre-alignment process. Furthermore, a software routine was developed to conduct pre-alignment and initiate the optimization coupling process. In Fig. 5(a), red dots indicate the white light reflected from the grating couplers corresponding to the input/output of each optical circuit included in the SOI chip.

Once positioned using the front CCD image and compared with the chip layout, a specific point for coupling light was identified. Initially, to see the coupling spot onto the chip, a red laser was directed through the fiber array to the input grating coupling of the 1×1 optical circuits, establishing the starting point. This basic optical circuit is integrated into each photonic design on the chip to estimate coupling losses by comparing output/input power signals. Online images were utilized simultaneously to optimize the first alignment stage [refer to Fig. 5(a)]. Subsequently, we aimed to optimize the coupling of

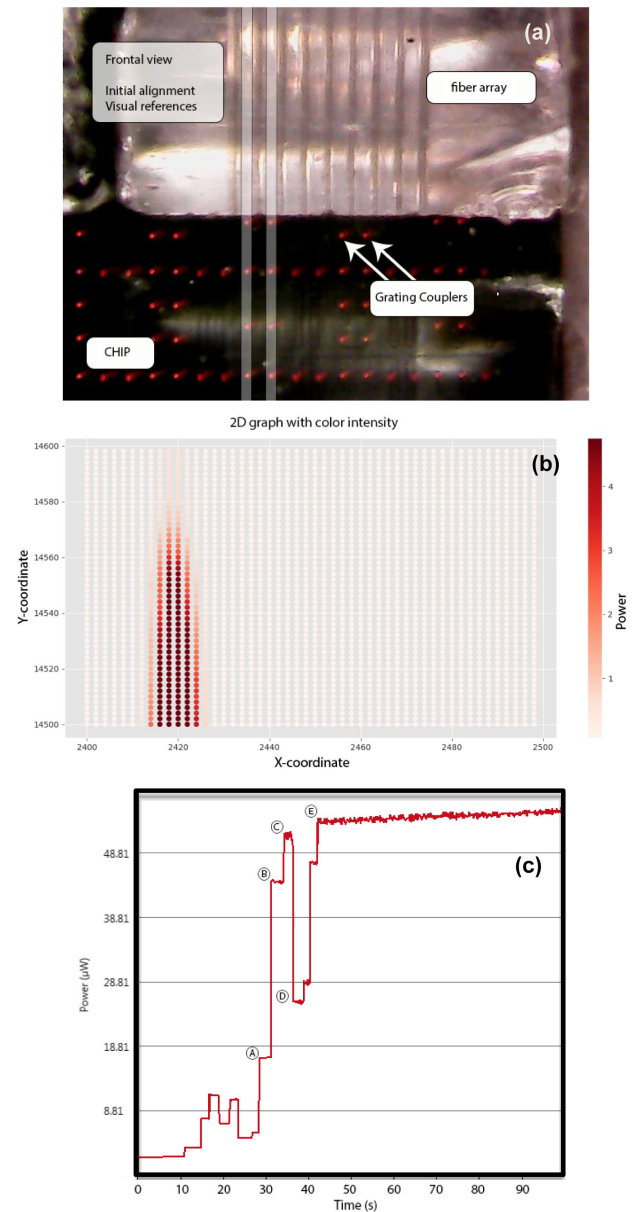


Fig. 5. (a) Image captured from the front microscope showing several illuminated light dots, corresponding to bright signals emanating from the grating couplers under white light illumination. The alignment with two gratings and two fibers from the fiber array is evident. The dotted lines indicate the strategy employed for this procedure. (b) Data representation following the pre-alignment routine, where a clear maximum dot is observed, indicating the optimal starting point for the experiments. (c) Graph depicting the output power as a function of time during the optimization of the point on the x and y plane within the explored zone (b).

laser light tuned to the optical communication band (approximately 1550 nm) into the chip. To initiate the pre-alignment process, a minimal optical signal from the selected integrated device (1×1) was required. This procedure involved executing multiple round trips around the starting point by scanning a dot matrix with a rectangular shape on the $x - y$ plane. The scan encompassed steps spaced $2 \mu\text{m}$ apart for each axis. The sample was moved in the X-Y plane while the fiber array remained stationary. Optical power received after coupling in different 1×1



Fig. 6. On the left is a photograph under UV illumination during the last step of the curing process. From this picture, we can also appreciate the shape and some details corresponding to the gluing process. As can be seen, a solid pedestal of adhesive is obtained, and from this it was possible to obtain a unique piece well established by the chip and fiber array. A picture taken during the UV curing process is presented on the right, which also shows below a mode from one naked fiber corresponding to one exit.

(input/output) devices was recorded using a digital/analog (D/A) converter from a LabJack card connected via USB to a PC. Data analysis identified the position yielding the highest signal, indicating optimal optical coupling. Figure 5(b) displays the data from following the pre-alignment procedure, showing that the power distribution can guide the most suitable light coupling conditions in photonic integrated structures. The black dots indicate the best coupling points obtained through this alignment strategy. Following the pre-alignment procedure, the chip was positioned at the optimal points within the explored region, and manual adjustments were made to enhance the output from the optical circuit under examination.

Figure 5(c) illustrates the additional optimization process, controlled with minimal x and y increments on the motorized station. The movement proceeded from left to right (x -axis) and top to bottom (y -axis), with a return to previous points when necessary. The figure highlights several relevant points labeled A, B, C, and D, detailing output power changes due to x and y control shifting. The up-down signals reflect the strategy of lateral and vertical movements to optimize output power, with point D representing a power plateau, indicating stable coupled light in the photonic system. The power difference between points C and D is attributed to mechanical system hysteresis.

Subsequently, an additional procedure was implemented to enhance the 1×1 optical circuit's output signal by incrementally adjusting the angular position of the fiber array, controlling minimal steps for each rotation angle. This adjustment aimed at lateral shifts due to the grating's broader angular acceptance from the elevation angle. Minor shifts were also applied to the lateral angles and the z -axis. The increased power registered during this final routine could produce a significant improvement, up to a factor of two compared to the initial value.

Conducting the optimization steps necessitated a detector change, requiring a more sensitive exploration to increase output signals from the optical circuit analyzed, thus employing a Germanium power meter (from Thorlabs Inc., USA).

2. Coupling Optimization and Assembling Chip-Fiber Array and Online Testing

After optimizing and completing the alignment for the 1×1 optical circuit, the testing of each 1×8 power splitter in the same section of the photonics chip was conducted. The realignment procedure was initiated for each 1×8 splitter, followed by fine manual adjustments using the control software of the six-axis motorized station. This involved making minor position adjustments to the chip-array fiber system, including small changes on the Z axis and angular positions in the X and Y planes of the photonic chip. This process resulted in minor shifts in elevation and lateral angles, leading to optical signal increases two or three times higher than the original. The measured data for the 1×8 SCC power splitter is presented in Table 1.

The next step involved assembling the chip-fiber array system into a robust unit for manipulation during packaging. This assembly was achieved by gluing the tandem using aerospace adhesive, NOA 68 from Norland (USA), a liquid photopolymer cured under ultraviolet light with wavelengths below 400 nm. Exposure to UV light for over 20 h is required; in this case, two 1-W LEDs and a 100-mW laser were used for 24 h. Upon full curing, good adhesion and solvent resistance were observed. Optimal adhesion is also obtained by aging at 50°C for 12 h, as shown in Fig. 6.

To carefully control the curing process of the adhesive, the output optical signal detected by the germanium power meter (from Thorlabs Inc., USA) was monitored in real-time. Additionally, the other optical exits from the system were measured directly from bare fiber optics using the beam profiler analyzer from Newport Inc. (USA), shown to the right of Fig. 6. The power for every exit from the splitter was analyzed once the curing process was fully completed.

Table 1 summarizes the average values of excess loss for the 1×8 power splitter developed in this work. Measurements were conducted both before and 24 h after the initiation of the gluing process, utilizing a UV-cured adhesive at a controlled temperature of 24°C . Additionally, results from simulations and experimental data reported in the literature are included for comparison.

Table 1. 1×8 Power Splitter Excess Loss before and after Assembly and UV-Curing Processes

1×8 Power Splitter (SOI)	Excess Loss (dB)	Coupling Method
Our splitter before UV curing (average)	5.8 (± 0.8)	Grating coupling
Our splitter after UV curing (average)	4.2 (± 0.5)	Grating coupling
Our simulations (average)	1.5 (± 0.5)	—
Reference [19]	7.6	End fire
Reference [20]	1.28	End fire
Reference [21]	6.74	Tapered fiber

As observed in Table 1, the excess loss was slightly reduced (approximately 1.6 dB lower) in the glued system. This reduction is attributed to the improved optical coupling facilitated by the adhesive, which matches the refractive index between the chip and the fiber array. The elimination of the air layer, which would otherwise increase the numerical aperture at the SOI-air interface, results in reduced light uncoupling from the chip. The results obtained are comparable to those reported for power splitters fabricated on SOI using various schemes, such as Y-branch and MMI [19–21]. Furthermore, numerical simulations indicate that the excess loss is approximately 1.5 dB. The discrepancy between experimental findings and simulations can be explained by two main factors: 1) the polarization of laser light for coupling to the splitters was not considered, and 2) the excess loss could also be attributed to imperfections in the fabrication process.

B. Packaging (Design, Assembling, and Install I/O Interfaces)

Considering the intended application of the power splitter developed in this work, a suitable case was designed for packaging the system, making it fit for installation within FTTH networks. To achieve a rapid and cost-effective prototype, 3D printing was employed using PLA materials. The packaging primarily includes internal clamps and coils to wind the acrylate-covered optical fiber, along with holders for fast optical fiber connectors (SC-UPC). Figure 7 illustrates the scheme of the designed case for the optical splitter. The coils were designed to allow easy winding of optical fiber paths, accommodating a large fiber diameter (around 10 cm) to minimize bending losses and birefringence effects. The package design also includes holders for the fiber connectors and screw sockets to secure the case.

Given the top-side light coupling and the fiber coiling process within the case, as well as to optimize and streamline assembly, the glued chip-fiber array system was positioned on the left side of the case. The system provides one optical input and seven optical outputs, compatible with SC/UPC connectors, to meet the requirements related to I/O interfaces.

Figure 8 presents the fabricated case for the integrated power splitter system, shown in an open position. It details the location of the chip and the coiling of the optical fiber. Additionally, the SC-UPC plugs are appropriately positioned within the designed packaging.

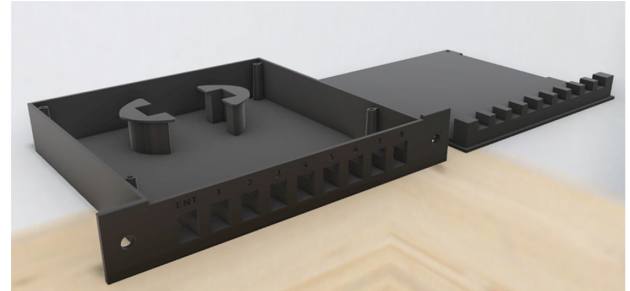


Fig. 7. The main details correspond to the inside view of the case. This image shows the final packaging design for the 1×8 power splitter as an open case. It shows the coils as well as the holder for the I/O optical interfaces.



Fig. 8. This image shows the open prototype of the power splitter fabricated and integrated in this work. The details of the case design are provided. As can be seen, the locations of the chip, fiber array (FA), internal clamps, screw sockets, coils for winding the fiber, and the fast connectors (I/O interfaces) are highlighted.

4. CONCLUSIONS

This paper introduces a straightforward, efficient, and cost-effective method for assembling a 1×8 power splitter fabricated on a silicon-on-insulator (SOI) platform. We provide a comprehensive description of the process that encompasses design, fabrication, assembly, and testing of the proposed power splitter. Particular emphasis is placed on the meticulous, step-by-step assembly process, which involves precise pre-alignment and fine adjustments to enhance output power.

For packaging, we utilized 3D printing to rapidly and economically produce a functional optical prototype. The

resulting photonic system demonstrates an excess loss of approximately 4.2 dB, making it well-suited for telecommunications applications in fiber-to-the-home (FTTH) networks. This performance is noteworthy given the packaging technology employed, and the expedited process used to realize the device.

Furthermore, this paper outlines the detailed assembly and packaging process for the 1×8 power splitter developed on a silicon photonics platform, ready for operation in FTTH networks. The innovative design and assembly/packaging procedures presented herein offer accessible implementation strategies that can be executed in low-complexity laboratories, making them ideal for faculty-student projects.

Funding. Agencia Nacional de Promoción Científica y Tecnológica (PICT-2016-4086, PICT-2017-0017); Universidad Nacional de Quilmes (PPROF-901-2018, PROF-827-947/20, PPROF-1329-2023).

Acknowledgment. JMV, RDS, and G. A. T thank CONICET and Universidad Nacional de Quilmes for supporting them. JMR also thanks Universidad Nacional de Quilmes for supporting his studies in the Automatic and Industrial Control Engineer degree.

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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